

HACKFEST 2026

Problem Statement Documentation & Technical Submission

Team ID	KH018
Team Name	Code Atlas
Team Leader Name	Sagar Yadav
Problem Statement No.	NA
Problem Statement Name	Green & Inclusive Travel — Smart Sustainable and Accessible Hospitality
Date of Submission	11 September 2026

SECTION 1 — TEAM DETAILS

Team ID	KH018
Team Name	Code Atlas
Team Leader Name	Sagar Yadav
Team Members Name(s)	Sagar Yadav (Team Leader) Yashraj Gadilkar Aditya Kadam Chinmay Gaidhane
Problem Statement Number	NA
Problem Statement Name	Green & Inclusive Travel — Smart Sustainable and Accessible Hospitality

SECTION 2 — UNDERSTANDING OF THE PROBLEM

Problem Statement

Planning a trip usually means searching through several different services. A traveler may need one platform for transport, another for hotels, another for tourist attractions, and additional sources for weather or environmental information.

The difficulty is not only finding these options, but choosing between them. A traveler may want a journey that is inexpensive, quick, accessible, and environmentally responsible at the same time. Existing travel searches often make these decisions separately, leaving the user to manually compare the results.

EcoTrail addresses this problem by bringing these considerations into one travel-planning system.

Problem Analysis / Background

A travel plan can depend on several conditions at the same time: budget, travel time, transport, accommodation, accessibility, environmental impact, and current conditions. The required information is often distributed across different services, so users have to manually search and compare it.

For example: "I want to travel from Pune to Mumbai within ₹5,000." The system must identify the origin, destination, budget and other requirements. A more detailed request may also include accessible accommodation and lower-carbon travel preferences.

Therefore, EcoTrail focuses on combining these requirements and using them together when selecting travel options.

Proposed Solution

EcoTrail is a travel-planning application where users can describe their requirements using normal conversational language. The system processes the request, identifies the trip requirements, collects relevant information from configured external services, evaluates the available options, and returns a personalized recommendation.

Workflow: User Request → Understand Trip Requirements → Collect Relevant Data → Organize & Normalize Results → Apply User Constraints → Evaluate Cost, Time, Accessibility & Emissions → Rank Suitable Options → Generate Travel Recommendation

The application will use Django and Django REST Framework for the backend and PostgreSQL for persistent data storage. AI/ML components will assist with understanding user requests, matching requirements, and producing useful recommendations.

Core Objective: EcoTrail aims to make travel planning a multi-factor decision rather than a simple search for the cheapest or fastest option, while helping users consider sustainability and accessibility alongside normal travel requirements.

SECTION 3 — RESEARCH

Research Conducted

During development planning, the team studied sustainable and inclusive travel planning and the technologies needed to implement the proposed system.

Research areas included: sustainable travel and carbon-emission comparison; accessible travel and accessibility information; travel recommendation systems; natural-language travel planning; transportation, accommodation, maps, places and weather APIs; AI/NLP approaches for requirement extraction and matching; and React, Vite, Django, Django REST Framework, PostgreSQL and Redis.

Existing Solutions / Similar Work

Existing travel platforms provide individual services such as transport search, hotel booking, maps, weather information and destination discovery. EcoTrail focuses on combining these factors into one recommendation workflow based on the user's complete requirements.

EcoTrail's intended distinction: Natural-language interaction + Multi-source data + Accessibility + Carbon estimation + Personalized ranking.

EcoTrail is not intended to replace existing travel services. It acts as an intelligent layer that brings relevant information together and helps users compare choices according to their requirements.

References / Sources

Official technical documentation and data references consulted for EcoTrail:

1. Django Documentation
2. Django REST Framework
3. PostgreSQL Documentation
4. Sentence Transformers Documentation
5. OpenStreetMap / Overpass API
6. Open-Meteo API Documentation
7. OpenRouteService Documentation
8. GHG Emission Conversion Factors (GOV.UK)
9. EcoTrail GitHub Repository

SECTION 4 — TECHNICAL DOCUMENTATION

Technology Stack

Frontend: React + Vite + JavaScript — user interface, chat interaction, trip/recommendation display, profile/preferences and authentication.

Backend: Django + Django REST Framework — authentication, trip management, request processing, external API integration, recommendation processing, carbon calculation and AI/ML integration.

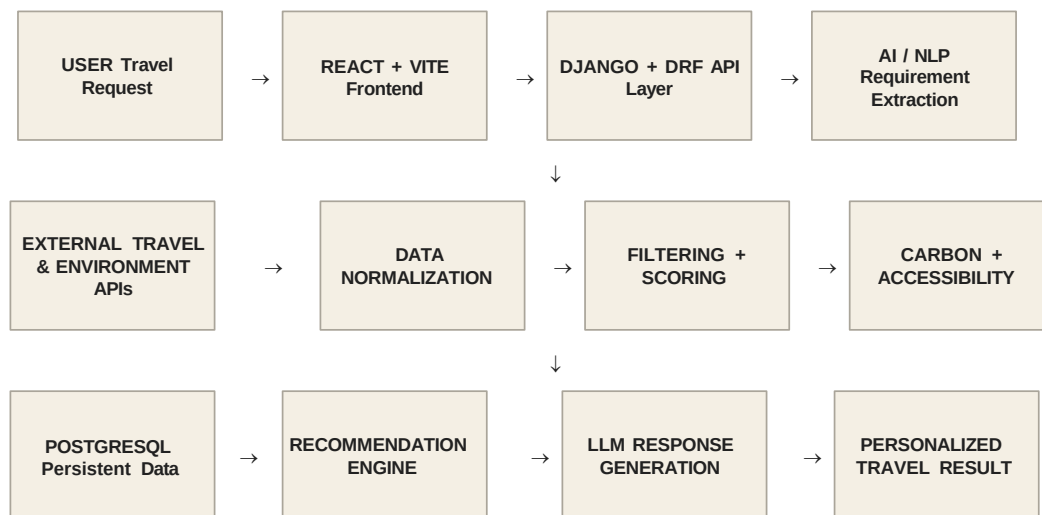
Database: PostgreSQL — users, preferences, trips, travel/accommodation data, accessibility information, chat sessions and carbon reference data.

Cache: Redis may be used for frequently requested information and repeated API results.

AI/ML: LLM/Generative AI, NLP, semantic matching and, if finalized, vision-based accessibility analysis.

External Services: transportation, accommodation, maps/routing, places, weather and environmental/carbon-related services.

Technical Flow / System Architecture Diagram



Working / Implementation Approach

1. The user enters a travel request in natural language.
2. AI/NLP identifies source, destination, dates, budget and preferences.
3. The backend requests relevant information from configured external services.
4. Results are normalized into comparable fields.
5. Mandatory constraints remove unsuitable options.
6. Remaining options are evaluated using cost, time, accessibility, carbon impact and user preferences.
7. Options are ranked and passed to the response layer.
8. The final recommendation is returned through the frontend.

AI/ML Details

EcoTrail uses AI to understand natural-language travel requests and extract important requirements such as destination, budget, duration, accessibility preferences and sustainability preferences. Semantic matching helps compare user requirements with available travel options, while AI-assisted response generation provides personalized recommendations.

Carbon impact is estimated using travel distance and an appropriate emission factor. The recommendation process considers factors such as cost, travel time, accessibility and estimated emissions. No custom-trained ML model is claimed unless it is developed and evaluated by the team.

SECTION 5 — ORIGINALITY & PLAGIARISM DECLARATION

- ✓ I confirm that the solution submitted by our team is developed during the hackathon. We have not submitted a pre-built project as our primary solution. Any open-source code, APIs, datasets, libraries, templates, or AI-assisted development used by us have been appropriately acknowledged.

Open-Source / External Resources Used

EcoTrail uses open-source frameworks and libraries including React, Vite, Django, Django REST Framework and PostgreSQL-related libraries. The project also uses external APIs and open data sources for maps, routing, weather and travel-related information. These resources are integrated to support the application's travel search, recommendation and sustainability features.

AI Tools Used

AI-assisted tools were used during development for technical research, code assistance, debugging, architecture planning and documentation. The generated suggestions were reviewed, modified and integrated by the team according to the project's requirements.

SECTION 6 — UPLOADS / ATTACHMENTS CHECKLIST

- Technical Flow / Architecture Diagram
- Research / Reference Document (if applicable)
- Project Documentation / Technical Report
- GitHub Repository Link
- Screenshots of Working Prototype
- Dataset / Sample Data (if applicable)
- Additional Diagrams (if any)

Team Leader Signature:  _____ Date: 11 September 2026